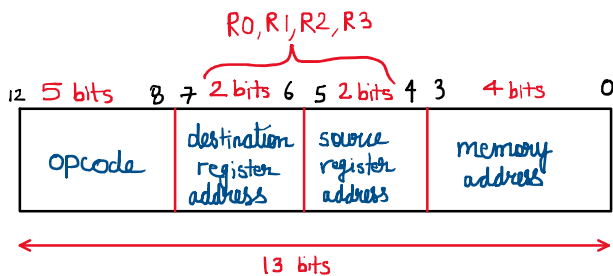


Mark-II

Improvements from Mark-I

- ① Harvard Architecture
- ② 8-bit datapath
- ③ Implement multiplication and division
- ④ Flag-based conditional branching
- ⑤ FPGA implementation

General Instruction Format



"The 4-bit field is either used as data memory address when required, or is unused/reserved and Program Counter (PC) selects instruction memory-location"

→ Datapath = 8-bits

→ Source register (R_s) = 2-bits (each stores 8-bits)

→ Destination register (R_d) = 2-bits (each stores 8-bits)

∴ Total register memory = $8 \times 4 = 32$ bits

→ Memory address = 4-bits (16 addressable memory locations)
(Harvard Arch.)

Instruction memory
(16×13 -bit)

Data Memory
(16×8 -bit)

Instruction Set Architecture (ISA) (23 of 32 used)

00000 — AND R_d, R_s
00001 — OR R_d, R_s
00010 — NOT R_d
00011 — XOR R_d, R_s
00100 — ADD R_d, R_s
00101 — SUB R_d, R_s
00110 — SLL R_d
00111 — SRL R_d

01000 — SEQ R_d, R_s
01001 — SLT R_d, R_s
01010 — SGT R_d, R_s
01011 — INC R_d
01100 — LOAD
01101 — STORE
01110 — JUMP
01111 — HALT

10000 — MUL

10001 — DIV

10010 — SRA

10011 — BEQ (Branch if equal)

10100 — BNE (Branch if not equal)

10101 — BC (Branch if carry)

10110 — BN (Branch if negative)

(23-31) — RESERVED

Categories

- ① Arithmetic — ADD, SUB, MUL, DIV, INC
- ② Logical — AND, OR, XOR, NOT
- ③ Shift — SLL, SRL, SRA
- ④ Comparison — SLT, SEQ, SGT
- ⑤ Memory — LOAD, STORE
- ⑥ Control — JUMP, HALT, BEQ, BNE, BC, BN

Flags

- ① Zero (Z) (for BEQ, $Z=1$; BNE, $Z=0$)
- ② Negative (N) (for BN, $N=1$)
- ③ Carry (C) (for BC, $C=1$)
- ④ Overflow (V)

Individual Instruction Formats

① Register Format : ADD, SUB, MUL, DIV, AND, OR, XOR, SLT, SEQ, SGT

② Single-Register Format : NOT, INC, SLL, SRL, SRA

③ Memory Format : LOAD, STORE

④ Control Format : JUMP, HALT, BEQ, BNE, BC, BN

OP	R_d	R_s	Memory
5-bits	2-bits	2-bits	4-bits

OP	R_d	R_s	Memory
5-bits	2-bits	(unused 6-bits)	

OP	R_d	R_s	Memory
5-bits	2-bits	2-bits	4-bits

OP	R_d	R_s	Memory
5-bits	(unused 4-bits)		4-bits